



Zilin Pan

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SUMMARY

Second-year Master's student focused on Embodied AI and Video Generation. Experienced in video-model evaluation, large-scale robotic data curation, multimodal training, and video-based action modeling. Interested in bridging generative world models and downstream control to translate visual understanding into reliable, executable actions for real-world robotics.

EDUCATION

School of Electronic and Computer Engineering, Peking University
Master of Computer Science and Technology

Sep. 2024 – Jun. 2027

School of Intelligent Systems Engineering, Sun Yat-sen University
Bachelor of Intelligent Science and Technology

Sep. 2020 – Jun. 2024

RESEARCH INTERESTS

Video Modeling : Video generation and evaluation for embodied intelligence.

VLA & WAM : Architecture optimization, multimodal training, and inference acceleration.

PUBLICATIONS

- Y. Deng*, **Z. Pan***, H. Zhang, X. Li, H. Qing, Y. Ding, Y. Zou, R. Wang, Y. Zeng, D. Zhou[†]. *Rethinking Video Foundation Model for the Embodied World*. *International Conference on Machine Learning (ICML)*, 2026 (Co-first author).
- H. Zhang*, Y. Deng*, **Z. Pan**, P.-T. Jiang, B. Li, Q. Hou, Z. Dou, Z. Dong, D. Zhou[†]. *TS-Attn: Temporal-wise Separable Attention for Multi-Event Video Generation*. *International Conference on Learning Representations (ICLR)*, 2026.
- P. Zhang*, Y. Deng*, S. Sun*, J. Ma, D. Wang, J. Du, **Z. Pan**, Y. Huang, H. Liang, S. Huang, R. Zhang, E. Xie, M.-Y. Liu, D. Zhou. *PhysisForcing: Physics Reinforced World Simulator for Robotic Manipulation*. *arXiv preprint arXiv:2606.28128*, 2026.
- Y. Deng*, **Z. Pan***, H. Zhang, X. Li, H. Qing, Y. Ding, Y. Zou, R. Wang, Y. Zeng, D. Zhou[†]. *Rethinking Video Foundation Model for the Embodied World*. *International Conference on Learning Representations (ICLR) Workshop*, 2026 (Co-first author).
- E. Zhang*, **Z. Pan***, Z. Yao, T. Dong, G. Chen, T. Deng, S. Chen, C.Y.-C. Chen[†]. Pre-trained language models for protein and molecular design. *Physical Chemistry Chemical Physics*, 2025, 27: 14189. (Co-first author)
- Z. Qiu, Z. He, Z. Zhan, **Z. Pan**, X. Xian, Z. Jin[†]. *SC-NAFSSR: Perceptual-Oriented Stereo Image Super-Resolution Using Stereo Consistency Guided NAFSSR*. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2023.

INTERNSHIP & RESEARCH EXPERIENCE

Huawei Internship — VLA Model Architecture and Inference Optimization

- Project:** Optimize VLA and WAM model architectures to improve multimodal training and inference efficiency. Systematically evaluate linear attention across different modules, comparing computational cost, training stability, and inference speed.

DAGroup Research Experience — Embodied Video Model Evaluation and Robotic Data Curation

- Project:** Built an end-to-end evaluation pipeline for embodied intelligence, assessing 30+ mainstream video models across 4 robot categories and 5 task categories in terms of physical plausibility, task-execution fidelity, and cross-robot generalization; collected and curated a large-scale 4M+ robotic-video dataset to support model training.